

Introduction

The purpose of this project was to develop an image classification model using Google Teachable Machine to recognise eight types of packaging materials and container objects. The project applied computer vision techniques to classify objects based on visual features such as shape, texture, transparency, and material appearance. This type of classification has practical applications in areas such as recycling systems, waste sorting, and automated packaging recognition. However, the task is challenging because several classes share similar visual characteristics. For example, aluminium beverage cans and steel food cans both have reflective metallic surfaces, while glass bottles and glass jars are visually similar due to their transparency and shape. The project therefore aimed to investigate how dataset refinement and training adjustments across three model versions could improve classification performance and model generalisation.

Dataset Construction

There was a total of 8 classes. Images were collected from Wikimedia, Google images and personal photographs captured on iPhone. There were two sets of images collected, the first been the development data set and the external dataset with fixed number (10 images) for each class making it a total of 80 images. It was ensured these images were different. During the first version 65 images had been gathered for each class making it a total of 520 images. For the second version 10 additional mages that were challenging had been added to each class (75) making the total sample 600 images. For the last version classes such as glass bottle, paper cup and aluminium beverage can got worse, thus additional 10 images had been added to those classes making the sample a total of 630 images. Images were selected for each class and placed into different folders. To ensure different images had been selected for each class, different variations were selected to make sure images had different lighting, background, angle (top shot, bottom shot), sizes and arrangements.

Model Development Details

For version 1, a total of 520 were used with 65 images per class. The Epoch was set to 45, the Batch size was 32 and the learning rate was 0.005. The model went through the entire data 45 times, while the data had been split to batch of 32 which meant the data would be grouped by 520/32 which is equivalent to 16 when all the 16 batches had been through the model then 1 batch is complete.

For the baseline (version 1) setting the learning rate to 0.005 helped the model learn quicker in the initial stages of the model development. This means the model makes bigger changes when it sees mistakes, so it can pick up patterns quicker. This was seen as a good starting point as it gives the results faster and shows how the model performs at the start. The results show that the model was able to recognise some packaging classes well, but performance was inconsistent across visually similar classes.

In Version 2 and Version 3, the learning rate was reduced to make learning more stable and improve the model accuracy. For the second version, to improve the dataset, harder images were added to each class making the sample a total of 600 images. For the class, clear plastic bottle, some of the similar images were swapped out to make sure the class had variety of images. The Epoch was set to 70, the Batch size had been reduced to 16 and the learning rate was also reduced to 0.001 this improved the overall stability in number of categories which was evident in classes clear plastic bottle, opaque plastic bottle, steel food can and aluminium beverage can.

For the version 3, the training parameters were as follows, where the epochs was increased to 100, while the Batch size was kept at 16 and the learning rate was reduced to 0.0005. For the classes that did not improve (glass bottle, paper cup and aluminium beverage can) additional images were added making the total sample 630 images. Version 3 produced the highest and most balanced performance across almost all classes. The increase in training epochs and the lower learning rate allowed the model to learn more carefully and improve feature recognition, especially in classes that were weak in earlier versions. The strongest improvements were, aluminium beverage can.

Sample Photos From Google Teachable Machine

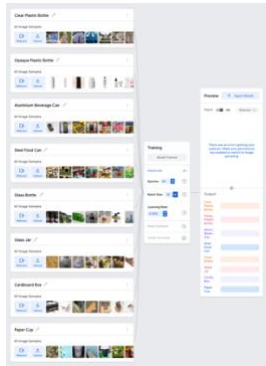


Figure 1: Version 1

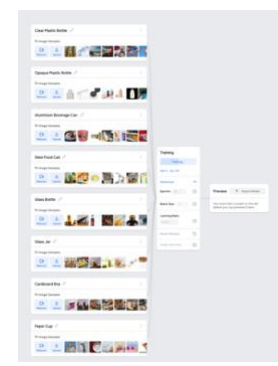


Figure 2: Version 2

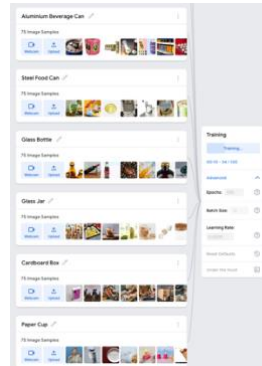


Figure 3: Version 3

Testing and Quantitative Results

Table 1: Performance Comparison across versions

Class	Version 1	Version 2	Version 3
Clear Plastic Bottle	67%	80%	83%
Opaque Plastic Bottle	86%	90%	96%
Aluminium Beverage Can	44%	57%	81%
Steel Food Can	70%	76%	90%
Glass Bottle	81%	71%	82%
Glass Jar	66%	70%	75%
Cardboard Box	87%	82%	88%
Paper Cup	82%	80%	89%

Sample screen Capture from Outputs of correctly identified Images:

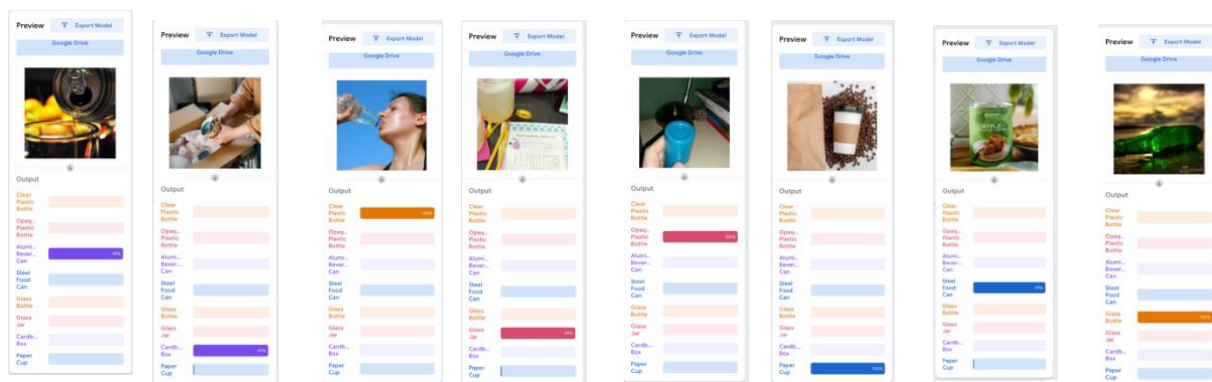
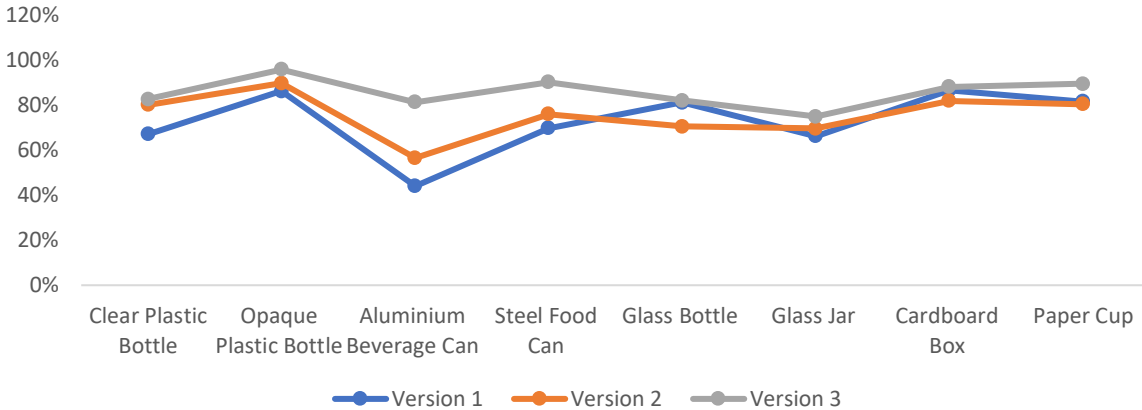


Figure 3: Correctly identified images with 100% accuracy rate across 8 classes

Graph 1: Performance Comparison of Classification Accuracy Across Model Versions



The first version, the model performed strongly on the following classes: cardboard box (87%), opaque plastic bottle (86%), paper cup (82%) and glass bottle (81%). This can mean these classes might have strong visual features and were easier for the model to learn. The weakest class was aluminium beverage can (44%). This suggests the model had difficulty learning features for aluminium cans, likely because they share visual similarities with steel food cans such as shape, reflective surface, and metallic texture. Other notable weak classes include, clear plastic bottle (67%) and glass jar (66%). These could have been hard to identify due to its transparency nature and similarity. Lastly, for the baseline model, inconsistency was recorded for clear plastic bottle as it showed unstable predictions of predicting some images confidently while others did not have a high confidence rate. To fix this issue and to ensure the model learns properly in other versions, un clear images had been removed and harder images had been added throughout.

For the second version, improvements had been noted throughout classes. For example, clear plastic bottle went from 67% to 80% (13% improvement), this suggests the added training images helped the model learn more variation in see through bottles. Another notable class which improved from 86% to 90% and remained strong was the Opaque Plastic bottle, the reason been the visual features are discrete compared to other classes. Other classes that improved include, steel food can improved from 70% to 76%, aluminium beverage can improved from 44% to 57% and lastly, glass jar improved from 66% to 70%. The following classes became weaker, glass went from 81% to 71%, this could mean the new images might have made the class boundaries less clear as the new training images introduced variations. Another notable class was the cardboard box, which went from 87% to 82% and lastly paper cup went from 82% with slight drop to 80%. For the second version, the model predictions still seem extreme for classes such as clear plastic bottle, glass bottle and paper cup. The model is not fully consistent as it performs very well on some images and vice versa on other images. This could mean some images are harder than others and there is visual overlap of classes. For the last version the three classes that did not improve had additional images added and the learning rate was further reduced to ensure consistent learning.

For the last version, aluminium beverage can had a significant improvements which went from V1 - 44% to V2 - 57% and V3 - 81%. This is the biggest improvement. It shows the model learned to separate aluminium cans better from steel food cans. This means Version 2's additional difficult images and Version 3's fine tuning worked well. Furthermore, steel food can also had strong improvement V1 - 70% to V2 - 76% and V3 - 90%. It shows better differentiation between metal containers. Lastly clear plastic bottle had steady improvement from V1 - 67% to V2 - 80% and V3 - 83%. This suggests the model

improved with the addition of different images. Lastly, the classes that struggled include glass jar from V1-66% to V2-70% and V3-75%. It has improved but it is lower than the other classes, this is likely the identical shape and material with glass bottle that went from V1-81% to V2-71% and V3-82%. For glass bottle, it recovered in version 3 as the training refinement helped.

Table 2: Version 3 External Test Results Per Class

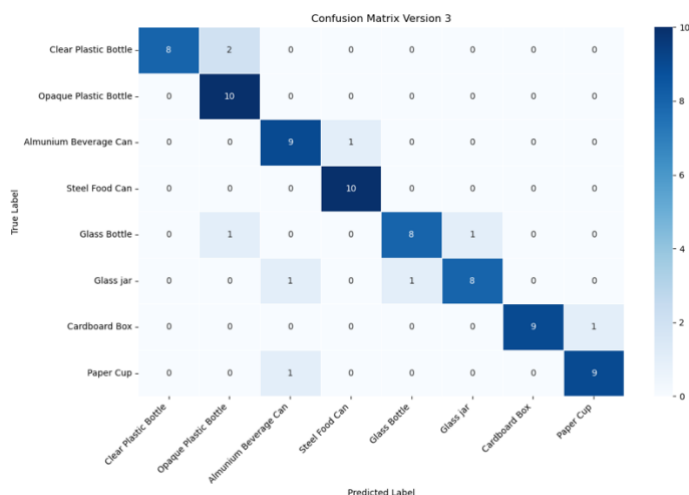
Class No	Class	Image	True label	Predicted label	Confidence score	Outcome
1	Clear Plastic Bottle	1	Clear Plastic Bottle	Opaque Plastic Bottle	0.91	In correct
		2	Clear Plastic Bottle	Clear Plastic Bottle	1	Correct
		3	Clear Plastic Bottle	Clear Plastic Bottle	1	Correct
		4	Clear Plastic Bottle	Clear Plastic Bottle	0.99	Correct
		5	Clear Plastic Bottle	Clear Plastic Bottle	0.79	Correct
		6	Clear Plastic Bottle	Opaque Plastic Bottle	0.47	In correct
		7	Clear Plastic Bottle	Clear Plastic Bottle	1	Correct
		8	Clear Plastic Bottle	Clear Plastic Bottle	0.98	Correct
		9	Clear Plastic Bottle	Clear Plastic Bottle	0.92	Correct
		10	Clear Plastic Bottle	Clear Plastic Bottle	1	Correct
2	Opaque Plastic Bottle	11	Opaque Plastic Bottle	Opaque Plastic Bottle	1	Correct
		12	Opaque Plastic Bottle	Opaque Plastic Bottle	1	Correct
		13	Opaque Plastic Bottle	Opaque Plastic Bottle	1	Correct
		14	Opaque Plastic Bottle	Opaque Plastic Bottle	0.99	Correct
		15	Opaque Plastic Bottle	Opaque Plastic Bottle	0.99	Correct
		16	Opaque Plastic Bottle	Opaque Plastic Bottle	0.59	Correct
		17	Opaque Plastic Bottle	Opaque Plastic Bottle	1	Correct
		18	Opaque Plastic Bottle	Opaque Plastic Bottle	1	Correct
		19	Opaque Plastic Bottle	Opaque Plastic Bottle	1	Correct
		20	Opaque Plastic Bottle	Opaque Plastic Bottle	1	Correct
3	Aluminium Beverage Can	21	Aluminium Beverage Can	Aluminium Beverage Can	0.87	Correct
		22	Aluminium Beverage Can	Aluminium Beverage Can	1	Correct
		23	Aluminium Beverage Can	Aluminium Beverage Can	1	Correct
		24	Aluminium Beverage Can	Steel Food Can	0.79	In correct
		25	Aluminium Beverage Can	Aluminium Beverage Can	0.65	Correct
		26	Aluminium Beverage Can	Aluminium Beverage Can	1	Correct
		27	Aluminium Beverage Can	Aluminium Beverage Can	0.8	Correct
		28	Aluminium Beverage Can	Aluminium Beverage Can	0.65	Correct
		29	Aluminium Beverage Can	Aluminium Beverage Can	0.99	Correct
		30	Aluminium Beverage Can	Aluminium Beverage Can	0.95	Correct
4	Steel Food Can	31	Steel Food Can	Steel Food Can	0.95	Correct
		32	Steel Food Can	Steel Food Can	0.67	Correct
		33	Steel Food Can	Steel Food Can	1	Correct
		34	Steel Food Can	Steel Food Can	0.99	Correct
		35	Steel Food Can	Steel Food Can	1	Correct
		36	Steel Food Can	Steel Food Can	0.94	Correct
		37	Steel Food Can	Steel Food Can	1	Correct
		38	Steel Food Can	Steel Food Can	1	Correct
		39	Steel Food Can	Steel Food Can	0.98	Correct
		40	Steel Food Can	Steel Food Can	0.49	Correct
5	Glass Bottle	41	Glass Bottle	Glass Bottle	1	Correct
		42	Glass Bottle	Glass Bottle	0.67	Correct
		43	Glass Bottle	Glass Bottle	0.96	Correct
		44	Glass Bottle	Glass Bottle	0.99	Correct
		45	Glass Bottle	Glass Bottle	1	Correct
		46	Glass Bottle	Glass Bottle	1	Correct
		47	Glass Bottle	Opaque Plastic Bottle	0.64	In correct
		48	Glass Bottle	Glass Jar	0.78	In correct
		49	Glass Bottle	Glass Bottle	1	Correct
		50	Glass Bottle	Glass Bottle	1	Correct
6	Glass jar	51	Glass jar	Glass jar	0.68	Correct
		52	Glass jar	Glass jar	0.91	Correct

		53	Glass jar	Glass jar	0.99	Correct
		54	Glass jar	Glass Bottle	0.41	In correct
		55	Glass jar	Glass jar	1	Correct
		56	Glass jar	Glass jar	0.98	Correct
		57	Glass jar	Glass jar	1	Correct
		58	Glass jar	Glass jar	0.89	Correct
		59	Glass jar	Glass jar	1	Correct
		60	Glass jar	Aluminium Beverage Can	0.87	In correct
7	Cardboard Box	61	Cardboard Box	Cardboard Box	1	Correct
		62	Cardboard Box	Cardboard Box	1	Correct
		63	Cardboard Box	Paper Cup	0.74	In correct
		64	Cardboard Box	Cardboard Box	0.95	Correct
		65	Cardboard Box	Cardboard Box	1	Correct
		66	Cardboard Box	Cardboard Box	1	Correct
		67	Cardboard Box	Cardboard Box	1	Correct
		68	Cardboard Box	Cardboard Box	1	Correct
8	Paper Cup	69	Cardboard Box	Cardboard Box	0.87	Correct
		70	Cardboard Box	Cardboard Box	0.98	Correct
		71	Paper Cup	Paper Cup	0.99	Correct
		72	Paper Cup	Paper Cup	1	Correct
		73	Paper Cup	Paper Cup	0.98	Correct
		74	Paper Cup	Paper Cup	0.98	Correct
		75	Paper Cup	Paper Cup	1	Correct
		76	Paper Cup	Paper Cup	0.93	Correct
		77	Paper Cup	Paper Cup	1	Correct
		78	Paper Cup	Aluminium Beverage Can	0.94	In correct
		79	Paper Cup	Paper Cup	1	Correct
		80	Paper Cup	Paper Cup	1	Correct

Looking at Table 2, The model performed well overall, with most of the images classified correctly. The main errors happened between visually similar classes, such as clear plastic bottle and opaque plastic bottle, aluminium beverage can and steel food can, and glass bottle and glass jar. For example, one clear plastic bottle image was predicted as opaque plastic bottle with confidence 0.91, and one glass jar image was predicted as aluminium beverage can with confidence 0.87. This shows the model was sometimes confused by similar shape, material, or transparency.

Confusion Matrix

Graph 2: Confusion matrix for Version 3



The heat map shows that Version 3 gets most classes right, with perfect scores for Opaque Plastic Bottle and Steel Food Can at 10 each, and strong results for Cardboard Box and Paper Cup at 9 each. Clear Plastic Bottle and Glass Bottle are slightly weaker, each with 8 correct results and a few mix-ups, mostly with similar container types. The main confusion appears between Glass Bottle and Glass Jar, and between Clear Plastic Bottle and Opaque Plastic Bottle, while Aluminium Beverage Can is mostly correct with 9 out of 10, but one item is mistaken for Steel Food Can.

Per Class Metrics for Version 3

Precision measures how often the model was correct when it predicted a class (Evidently AI, 2025). Recall measures how many actual images in a class were correctly identified (Evidently AI, 2025). The F1-score combines precision and recall into a single balanced measure (Great Learning Editorial Team, 2025). The macro-average F1-score is calculated by averaging the F1-scores across all classes (Jeevan, 2025).

Table 3: Per Class Precision, Recall, F1 Score for Version 3

Class	Precision	Recall	F1 Score
Clear Plastic Bottle	1.0000	0.8000	0.8889
Opaque plastic bottle	0.7692	1.0000	0.8696
Aluminium Beverage Can	0.8182	0.9000	0.8572
Steel Food Can	0.9091	1.0000	0.9524
Glass bottle	0.8889	0.8000	0.8421
Glass jar	0.8889	0.8000	0.8421
Cardboard box	1.0000	0.9000	0.9474
Paper cup	0.9000	0.9000	0.9000

An example calculation for **Clear Plastic Bottle** is shown below. Precision was calculated as True Positive divided by True Positive plus False Positive (Evidently AI, 2025). Recall was calculated as True Positive divided by True Positive plus False Negative (Evidently AI, 2025). F1-score was calculated using the harmonic mean of precision and recall (Great Learning Editorial Team, 2025).

- Precision = $8/(8+0) = 1.0000$
- Recall = $8/10 = 0.8000$
- F1-score = $2 * (1.000 * 0.8000) / (1.000 + 0.8000) = 0.8889$

The macro-average F1-score was then calculated by averaging the F1-scores across all eight classes (Jeevan, 2025).

There was strong performance on the external test dataset as the overall accuracy for the final model was 88.75% (71/80). For majority of the classes the precision scores were fairly high, with cardboard box and plastic bottle achieving 100% precision meaning all the predicted samples for these classes were correct. In terms of the recall scores, there was a perfect detection for steel food can and opaque plastic bottle. On the other hand, lower recall was reported for clear plastic bottle, glass bottle and glass jar which insinuates the model struggled with identifying transparent and visually similar objects. The F1-scores ranged from 0.8182 to 0.9524, showing a good balance between precision and recall across classes. Lastly, the macro-average F1-score was 0.8875, indicating strong overall classification performance across all eight classes. These results show the model generalised well, but transparent and visually similar packaging types remain the most challenging.

Analysis and Discussions

The most common classes model confused as they were similar include, glass bottle and glass jar, aluminium beverage can and steel food can. In some instances cardboard box was mixed up paper cup. Some of the notable limitations that should be considered include some of the images had plain background, although effort was made to ensure different lighting and arrangements of objects was used to diversify the images in the sample. Although, most of the classes (5/8) had same number of images while the remaining 3 classes had 10 extra images as they needed further refining for the last version. For the second version, 10 visually ambiguous images had been added to each class to

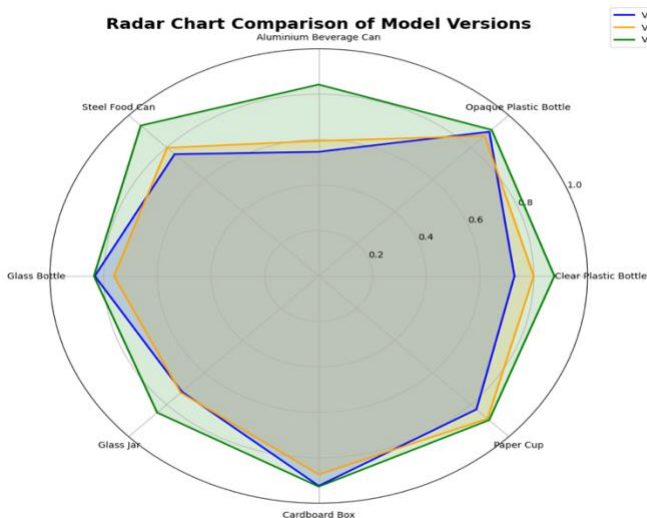
diversify the data and enhance model generalisation through exposing the model to more realistic and challenging object variations. Furthermore, in some classes, difficult images had increased class ambiguity leading to a momentary performance variabilities. The model did its best to be generalised to the external dataset. The model could be reliable in real world settings but with caution.

Table 4: Accuracy, Precision and recall of Version 1,2,3

Class	Version 1				Version 2				Version 3			
	Accuracy	Precision	Recall	F1	Accuracy	Precision	Recall	F1	Accuracy	Precision	Recall	F1
Clear plastic bottle	0.6700	0.7143	0.8000	0.7547	0.8000	0.8000	0.8000	0.8000	0.8300	1.0000	0.8000	0.8889
Opaque plastic bottle	0.8600	0.8333	1.0000	0.9091	0.9000	0.8182	0.9000	0.8571	0.9600	0.7692	1.0000	0.8696
Aluminium beverage can	0.4400	0.8000	0.4000	0.5333	0.5700	0.7143	0.5000	0.5882	0.8100	0.8182	0.9000	0.8571
Steel food can	0.7000	0.8750	0.7000	0.7778	0.7600	0.8333	0.8000	0.8163	0.9000	0.9091	1.0000	0.9524
Glass bottle	0.8100	0.8889	0.8000	0.8421	0.7100	0.875	0.7000	0.7778	0.8200	0.8889	0.8000	0.8421
Glass jar	0.6600	0.8000	0.7000	0.7467	0.7000	0.7778	0.7000	0.7368	0.7500	1.0000	0.8000	0.8889
Cardboard box	0.8700	1.0000	0.9000	0.9474	0.8200	0.9000	0.9000	0.9000	0.8800	1.0000	0.9000	0.9474
Paper cup	0.8200	0.7692	0.9000	0.8293	0.8000	0.9643	0.9000	0.9310	0.8900	0.9000	0.9000	0.9000

The table shows that Version 3 gives the strongest results overall. It reaches the highest scores in most classes, with especially good results for Opaque Plastic Bottle, Steel Food Can, Cardboard Box, and Paper Cup, while Version 1 is more uneven and drops sharply for Aluminium Beverage Can. Across the classes, Cardboard Box is one of the easiest to identify, with scores close to 0.87 to 0.90 in all three versions, and Paper Cup is also strong, especially in Version 2 and Version 3. The weakest class is Aluminium Beverage Can, where Version 1 only reaches 0.44 accuracy and 0.40 recall, while Version 3 improves it to 0.81 accuracy and 0.90 recall. Overall, the figures show that the model gets better with each version, and Version 3 is the most consistent across the main classes.

Graph 3: Radar Chart comparing Versions



The radar chart shows that Version 3 performs best overall across the main classes, with particularly strong scores for Steel Food Can (F1=0.95), Cardboard Box (F1=0.95), Glass Jar (F1= 0.89), Paper Cup (F1= 0.90), and Clear Plastic Bottle (about 0.86). Version 1 and Version 2 are slightly lower and more uneven, with both sitting around 0.70–0.90 in most classes. For example, Clear Plastic Bottle improves from about 0.75 in Version 1 to 0.80 in Version 2 and 0.89 in Version 3, while Steel Food Can increases from about 0.78 to 0.82 and then to 0.95. All three versions are strongest on Cardboard Box, where scores are close to 0.97–0.99, and the weakest area is Aluminium Beverage Can, where Version 1 is about 0.53, Version 2 about 0.59, and Version 3 improves to about 0.86.

Table 5: Overall performance across model versions

Version	Overall Accuracy	Macro Precision	Macro Recall	Macro F1
Version 1	77.50%	0.8017	0.775	0.7925
Version 2	81.25%	0.8354	0.8125	0.8009
Version 3	88.75%	0.8982	0.8875	0.8933

Table 5 shows a clear improvement across the three versions. Version 1 has an overall accuracy of 77.50%, Version 2 improves to 81.25%, and Version 3 performs best at 88.75%. The same pattern appears in the other figures: macro precision rises from 0.8017 in Version 1 to 0.8354 in Version 2 and 0.8982 in Version 3, while macro recall increases from 0.775 to 0.8125 and then 0.8875. Macro F1 also improves from 0.7925 in Version 1 to 0.8009 in Version 2 and 0.8933 in Version 3. In simple terms, each new version does better than the last, and Version 3 is the strongest across all measures.

Conclusion

This project demonstrated the use of computer vision for classifying different types of packaging objects across eight classes. Through three model versions, the results showed clear improvements in classification performance as the dataset was refined and the training settings were adjusted. The final model achieved the strongest and most balanced results, reaching an overall accuracy of 88.75% and a macro-average F1-score of 0.8875, which indicates strong overall performance across the classes.

The findings showed that increasing the number of training images, introducing more challenging examples, lowering the learning rate, and increasing the number of training epochs all contributed to better performance. Version 3 performed particularly well for classes such as opaque plastic bottle, steel food can, cardboard box, and paper cup. One of the biggest improvements was seen in aluminium beverage can, where the model improved significantly compared to the earlier versions.

Despite these improvements, some limitations remained. The model still struggled with transparent and visually similar objects, especially glass bottle and glass jar, as well as aluminium beverage can and steel food can. These errors highlight the difficulty of distinguishing objects that share similar shapes, textures, or reflective surfaces. Some inconsistencies may also have been caused by background similarities and image ambiguity within the dataset.

Overall, the project showed that careful dataset construction and gradual model refinement can significantly improve image classification performance. The final model demonstrated good generalisation on the external test dataset and could be useful in practical applications involving packaging recognition and sorting. However, further improvements could be made by increasing dataset diversity, using more real-world images, and exploring methods that improve recognition of visually similar and transparent objects.

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